Paleomagnetism of four late Miocene gabbroic plugs in south-central British Columbia

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Received March 11, 1969
Accepted for publication May 5, 1969

A paleomagnetic study was made of 101 cores from 20 sites representing about equally 4 small olivine gabbro plugs which intrude tectonically undisturbed olivine-rich plateau basalts of late Miocene age in the southern Cariboo region of south-central British Columbia. After alternating-field cleaning, statistical analysis of the stable remanence indicates that the site mean directions are significantly distinct within each plug so that even such small intrusive bodies (250–800 ft (78–248 m) in diameter) must be thoroughly sampled to derive a representative mean remanence direction. Both Tin Cup Mountain and Lone Butte plugs have normally polarized remanence, whereas Mount Begbie and Forestry Hill plugs have reversely polarized remanence. Statistical analysis indicates that these plugs were emplaced over a short span of geologic time of possibly less than 1×10^6 years and that at least three polarity intervals are represented. The geomagnetic pole position computed from the normalized site mean directions is 146.7° W, 84.9° N ($\delta p = 4.8°$, $\delta m = 5.5°$). This pole position is almost coincident with the pole position determined for the surrounding plateau basalts, and it is consistent with those obtained from other Miocene formations. These results support the hypothesis that the plugs represent the original volcanic vents from which the plateau basalts were extruded.

Introduction

At least four gabbroic plugs have intruded the plateau basalts in the Bonaparte River map-area (National Topographic System 92P) (Campbell and Tipper 1966, 1969) or south Cariboo region of the Interior Plateau of British Columbia (Fig. 1). Farquharson (1963) mapped the geology and described the petrology of two of these plugs—Mount Begbie and Forestry Hill—in considerable detail, and he gave brief notes on the other two—Tin Cup Mountain and Lone Butte.

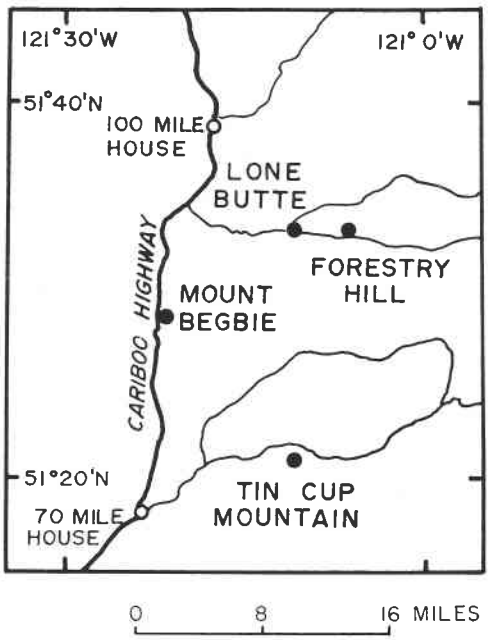


FIG. 1. Location map showing the sampled plugs.

The plugs stand 100 to 200 ft (30.5–61 m) above the surrounding plateau surface as elliptical knobs with a maximum diameter in plan view of 250 to 800 ft (76–244 m) (Fig. 2). They are composed of unaltered olivine gabbro with minor lenses of leucogabbro, picritic gabbro, dolerite, and pegmatitic gabbro derived from an alkali basaltic magma. Farquharson (1963) reported that the foliation given by the orientation of the feldspar crystals dips toward the center and strikes parallel to the adjacent wall for the two plugs that he investigated in detail. He attributed the foliation pattern to the upward flow of magma.

Mathews and Rouse (1963, p. 56) reported that the surrounding plateau basalts consist of "dark olivine basalts occurring in a series of thin . . . lava sheets" in which "regional dips rarely exceed 1°". Thus the plugs have apparently been unaffected by any tectonic rotation.

None of the plugs has been radiometrically dated. Mathews and Rouse (1963) used paleontological evidence derived from fossils in interbedded sediments to date the plateau basalts as middle Miocene to early Pliocene. Also, they reported three K–Ar dates from the base and middle of the lava sequence of 13 ± 2,

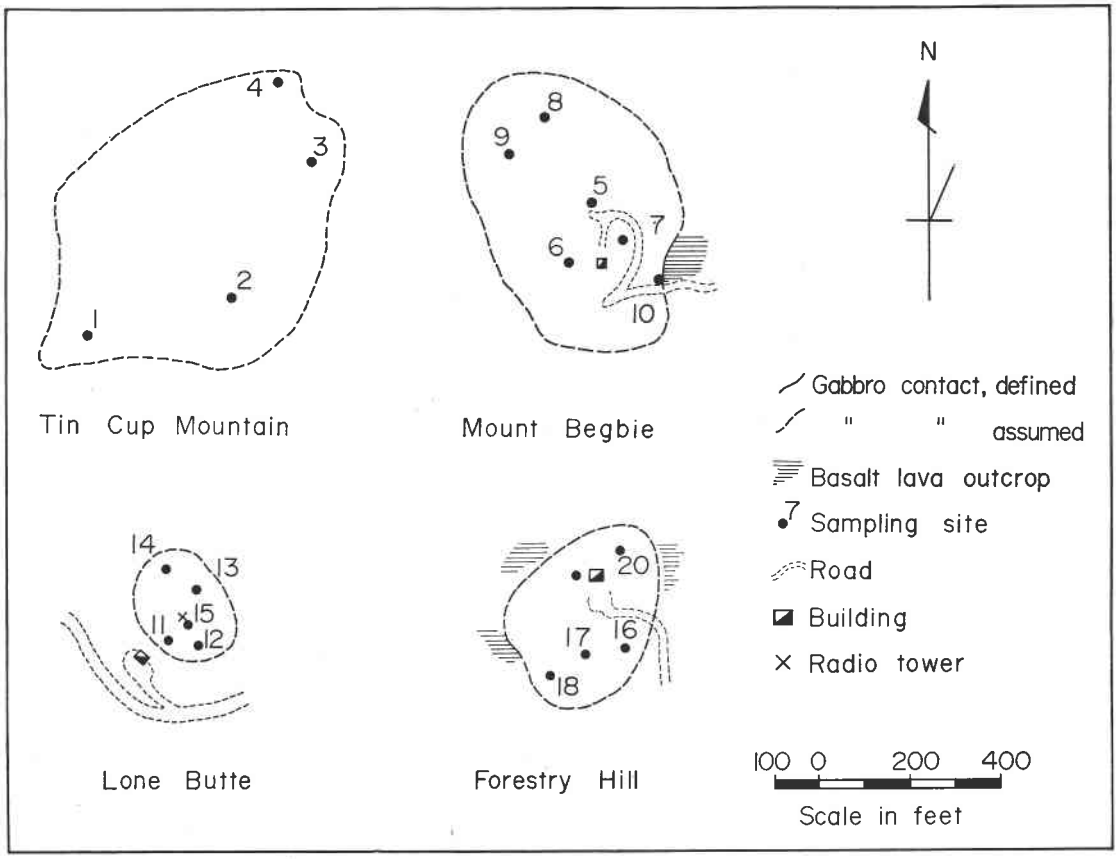


FIG. 2. Sampling sites showing each plug in plan view.

12 ± 2, and 10 ± 2 m.y. (see also Mathews 1963), which indicates a late Miocene age based on the Geological Society of London (1964) time scale. On the basis of the geological and petrological relationships between the plugs and flows, Farquharson (1963) postulated that the plugs occupy former volcanic vents, which fed lava to the surrounding flows.

Because the paleomagnetic method seemed adequate *a priori* to examine Farquharson's hypothesis, oriented samples were drilled from the four plugs in July 1968. The data obtained from these samples was examined to show that the four plugs are essentially comagmatic, and to provide a reliable pole position for the intrusive event. By comparison of this pole position to that derived from the surrounding lavas (Symons, in preparation), it will be shown that the plugs and lavas are probably comagmatic, which indicates that the plugs were intruded during late Miocene time.

Sampling

One hundred and one cores were drilled from 20 sites located on the four plugs as shown in Fig. 2. The sites were selected to give representative coverage of each plug. Generally five cores were drilled from each site, with the individual cores collected from points separated by several feet. The cores were oriented in situ using a solar compass with topographic and Brunton compass check orientations. Two cylinders, $1\frac{1}{4}$-inches (3.2 cm) in diameter and $1\frac{3}{32}$ inches in length, were cut from each core, yielding a total of 202 cylinders.

Measurement and Demagnetization of Remanence

The direction and intensity of the natural remanent magnetization (NRM) was measured for each cylinder using an automated biastatic magnetometer (Larochelle and Christie 1967).

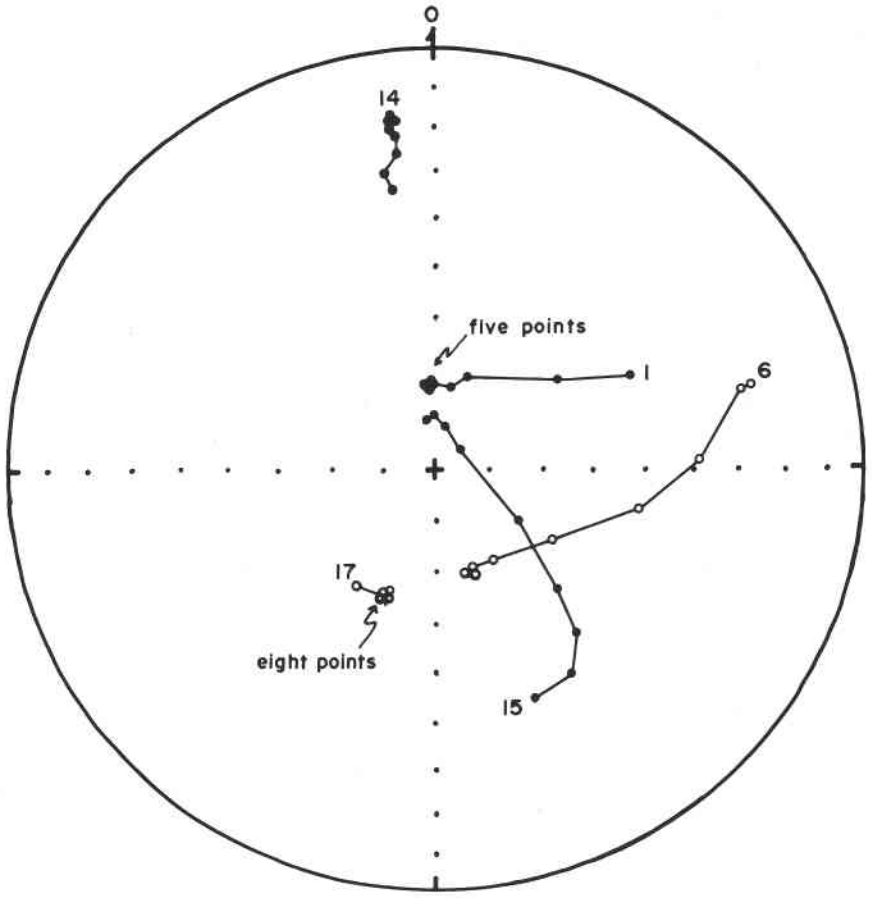


FIG. 3. Stereonet showing 5 examples of AF demagnetizing direction plots for test cylinders. The numbers indicate the site represented and NRM direction dot with successive unnumbered dots representing 25, 50, 75, 100, 200, 300, 400, and 800 Oe (peak) 'cleaning' fields. Solid (open) circles indicate positive (negative) or downwardly directed (upwardly directed) remanence vectors.

One test cylinder with an 'average' NRM direction and intensity was selected from each site to represent it. Using the alternating-field (AF) demagnetizing apparatus described by Larochelle and Black (1965), each test cylinder was demagnetized at successive peak field values of 25, 50, 75, 100, 200, 300, 400, and 800 Oe and its residual remanence measured after each step. The successive change in remanence direction (for which five examples are shown on Fig. 3) and intensity (Fig. 4) were plotted for each test cylinder. Also their stability index (SI) values (Tarling and Symons 1967) were calculated (Table I).

Examination of the remanence direction plots indicate that the test cylinders from sites: (a) 1, 2, 3, 4, 5, 9, 10, 11, and 17 reach apparent stability by 100 Oe (e.g., no. 1 and 17, Fig. 3) and yield an average SI of 27.9; (b) 6, 7, 8, 15, 16, 18, 19, and 20 reach apparent stability by 800 Oe (e.g., no. 6 and 15, Fig. 3), with no. 7 and 20 just reaching apparent stability and yield an average SI of 17.6; and (c) 12, 13, and 14 appear to approach but do not reach apparent stability by 800 Oe (e.g., no. 14, Fig. 3) and yield an average SI of 10.8. The change in remanence intensities (Fig. 4) confirms the directional data. For example, the test cylinders from sites 1 to 4 from Tin Cup Mountain plug show little change and are very similar in intensity by 100 Oe; and the cylinders from sites 7 and 8 have very intense NRM, but drop by 800 Oe demagnetization to the range of intensities found for the remaining sites (5, 6, 9, and

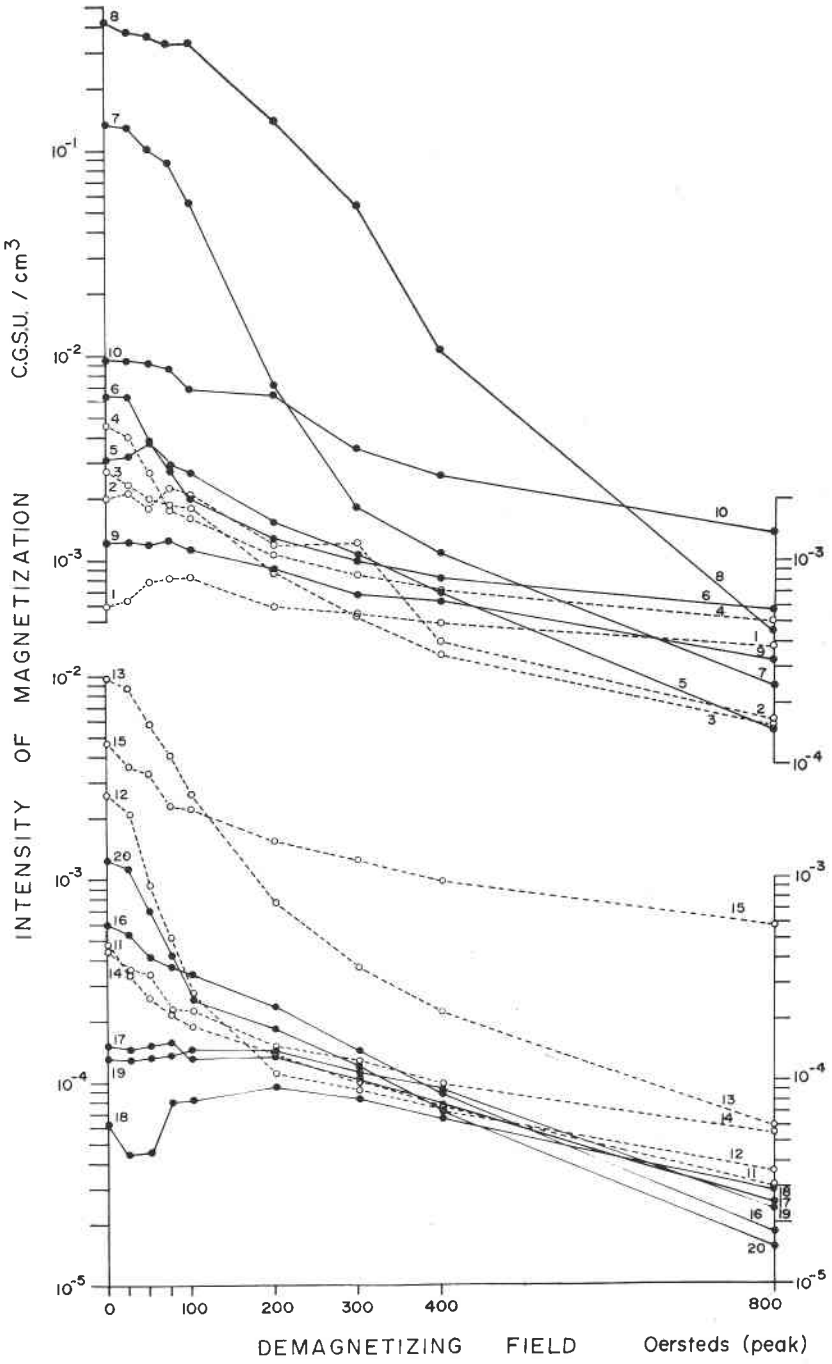


FIG. 4. AF demagnetizing intensity plots for test cylinders with the number indicating the site represented. Solid (open) circles indicate positive (negative) polarity for the site.

TABLE I
Remanence data by sites

Site No.	Int.	N	SI	Hmin	Hmax	H_{AF}	Reject	n	R	$D_s°$	$I_s°$	δ_{se}	θ_s
1	TC	5	39.6	300	800	300		5	4.9965	358.3	74.9	1.1	2.7
2	"	5	19.5	75	200	200	1b	4	3.9953	337.5	75.5	1.6	1.4
3	"	5	14.0	100	800	400	1b	4	3.9975	350.2	73.5	1.1	1.8
4	"	5	17.6	300	800	400	1b	4	3.9987	342.4	77.4	0.9	1.4
5	MB	5	23.6	200	400	200		5	4.9927	155.4	−71.5	1.6	3.8
6	"	5	42.9	300	800	400	1b	4	3.9894	151.5	−69.3	2.4	2.7
7	"	5	10.0	0	75	800	1b	4	3.9889	161.2	−71.0	2.5	4.4
8	"	5	21.4	25	400	800	1a, 1b	3	2.9710	156.3	−75.3	5.6	5.0
9	"	5	39.3	100	800	200		5	4.9940	184.2	−72.8	1.4	2.0
10	"	5*	60.8	25	800	300	1a	4	3.9790	162.9	−71.2	3.4	1.5
11	LB	5	16.5	200	800	300	1b	4	3.9909	339.2	66.7	2.2	2.8
12	"	5	10.4	300	800	400**		5	2.7172	46.6	69.6	28.4	3.6
13	"	6	10.2	300	800	400**	2a	4	3.7735	41.9	70.9	11.2	6.3
14	"	5	11.8	50	100	400**		5	4.1066	13.7	30.9	17.8	6.3
15	"	5	13.0	300	800	400		5	4.9613	15.2	72.6	3.7	5.7
16	FH	5	15.9	200	400	300	1b	4	3.9927	193.7	−61.9	2.0	2.3
17	"	5	20.4	300	800	200	1b	4	3.9922	193.0	−65.0	2.1	3.6
18	"	5	14.7	200	800	300	1b	4	3.9932	203.3	−60.6	2.0	2.6
19	"	5	15.7	300	800	300	1b	4	3.9971	189.1	−71.9	1.2	1.5
20	"	5	7.2	200	800	300	1a, 1b	3	2.9986	186.9	−69.6	1.2	3.1

Int = intrusive plug: Tin Cup Mountain (TC), Mount Begbie (MB), Lone Butte (LB), and Forestry Hill (FH).
N = number of cores collected at site.
SI = stability index.
Hmin(Hmax) = minimum (maximum) demagnetizing field of stability index range in oersteds (peak).
H_{AF} = demagnetizing field used to 'clean' site remanence in oersteds (peak).
Reject = rejected cores specified by number and reason: (a) angle between the remanence directions of the two cylinders exceeds 15°; (b) single core direction diverges from the mean remanence direction of the remaining cores by more than $3 \times \delta_s$.
n = number of cores used to calculate R, D_s, I_s, δ_{se}, and θ_s.
R = resultant giving each core unit weight.
$D_s(I_s)$ = declination (inclination) of site mean remanence vector in degrees.
δ_{se} = angular standard error in degrees.
θ_s = mean angular deviation between the remanence directions of the two cylinders of each core in degrees.
*includes 3 cores of completely altered basaltic lava at contact of which 1 core is rejected.
**similar data obtained after 800 Oe (peak) 'cleaning'.

10) from the Mount Begbie plug. From the remanence data obtained from the demagnetization of the test cylinders, a specific field (as shown on Table I) was selected for demagnetization of the remaining cylinders from each site. The remaining 182 cylinders were demagnetized and their residual remanence remeasured. The cylinders from sites 12, 13, and 14 were given a second AF 'cleaning' in 800 Oe to see if their remanence directions would cluster more tightly; however, they did not.

Data Selection

Two steps were taken to assure that the remanence data used for analysis were stable and homogeneous.

First, homogeneity in a core was considered satisfactory if the angle between the remanence directions of the two cylinders did not exceed 15°. Five cores proved inhomogeneous on this basis and were not considered further. They came from sites for which the AF demagnetizing test cylinders also yielded evidence of instability. The mean value (θ_s) for the θ angles for each site using homogeneous cores only is given in Table I. For all 20 sites the value of θ_s is $\leqslant 6.3°$, which indicates that most cores have stable homogeneous remanence. For four sites the value of θ_s is $\leqslant 1.5°$, which may be considered greater than the probable measurement error.

Second, the angular standard deviation (δ_s) was calculated for each site giving each homogeneous core unit weight. If the angular deviation of one core from the mean position of the remaining cores exceeded the value of ($3 \times \delta_s$) for the remaining cores, then the deviating core was considered an erratic and rejected. In this way possible gross sampling or measurement errors were eliminated. Twelve cores were rejected on this basis, and several of these had very intense NRM (i.e., 10^{-1} c.g.s.v./cm^3), which could be due to a lightning-induced anhysteric remanent magnetization, since the sites are topographically high compared with the surrounding plateau.

The resulting site mean remanence directions

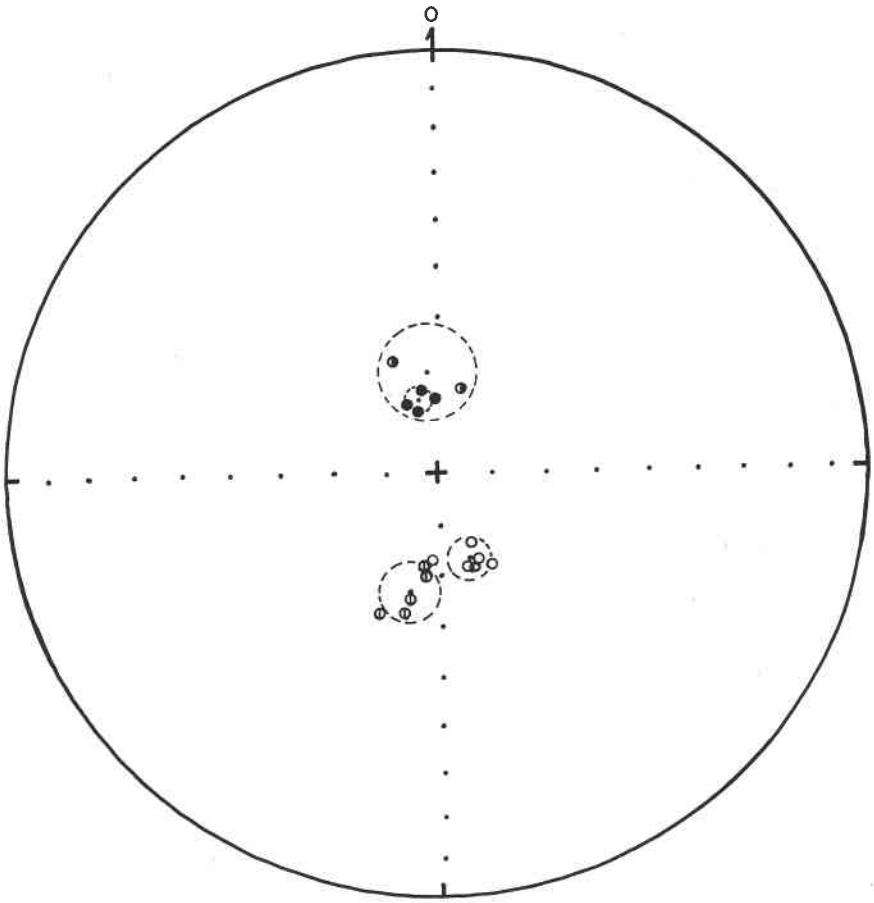


FIG. 5. Site mean remanence directions for Tin Cup Mountain (solid circles) and Lone Butte (half solid circles) plugs, which both have positive polarity, and for Mount Begbie (open circles) and Forestry Hill (crossed open circles) plugs, which both have negative polarity. The intrusive mean direction for each plug is indicated by a dot and its cone of 95% confidence by the dashed line giving each site unit weight.

in Table I are defined by their respective declination (D_s), inclination (I_s), and standard error (δ_{se}), which is the angular standard deviation (δ_s) divided by the square root of the number of observations (Larochelle 1968). Examination of δ_{se} for the 20 sites reveals that it is considerably smaller than 10° except for sites 12, 13, and 14 on Lone Butte, for which it is greater. This observation confirms the AF demagnetization results in that the test cylinders for these three sites did not reach apparent stability. The results from these three sites were not considered further in this analysis.

Statistical Analysis

A preliminary look at the site mean remanence directions (Fig. 5) shows that Tin Cup Mountain and Lone Butte plugs have positive northerly directions, whereas Mount Begbie and Forestry Hill plugs have negative southerly directions, which are almost diametrically opposed to those of Tin Cup Mountain and Lone Butte plugs.

For each intrusive plug, the mean remanence direction (D_i, I_i), the angular standard error (δ_i) and the radius of the cone of 95% confidence (α_{95}) were calculated giving each reliable site and each reliable core unit weight; also, the within-site angular variance (δ^2_w), the between-site angular variance (δ^2_b) and their ratio (δ^2_b/δ^2_w) were calculated following the method of Larochelle (1968) (Table II). The intrusive mean remanence direction and α_{95} for each plug are shown on Fig. 5. In all four plugs,

TABLE II

Summary of remanence data and angular variance ratio tests

Intrusive plugs (S)	B	N	R	$D_i°$	$I_i°$	δ_i	α_{95}	δ^2_b	δ^2_w	δ^2_b/δ^2_w	$F_{2(B-1),2(N-B),.05}$
Tin Cup Mountain	4		3.9964	347.4	75.5	1.4	2.4				
(TC)		17	16.9725	348.1	75.3	0.8	1.4	.01035	.00185	5.60	2.47
Mount Begbie	6		5.9871	161.6	−72.1	1.7	2.9				
(MB)		25	24.8582	162.4	−72.1	1.2	2.2	.02281	.00894	2.55	2.09
Lone Butte	2		1.9859	354.6	70.5	6.8	11.8				
(LB)		9	8.8900	356.6	70.9	3.2	5.5	.12514	.01365	9.17	3.74
Forestry Hill	5		4.9814	194.0	−65.9	2.5	4.3				
(FH)		19	18.9027	194.3	−65.7	1.4	2.4	.03560	.00374	9.51	2.29
		P_1		D_q	I_q	δ_q					$F_{2(P_1-1),2(B-P_1),.05}$
Normal plugs	6		5.9768	350.3	73.8	2.2	3.9				
(TC; LB)		2	1.9980	352.9	73.2	2.5	4.4	.01142	.00878	1.30	4.46
Reverse plugs	11		10.8993	178.6	−70.1	2.4	4.2				
(MB; FH)		2	1.9873	180.7	−69.6	6.4	11.2	.13894	.00699	19.87	3.55
		P_0		D_a	I_a	δ_a					$F_{2(P_0-1),2(B-P_0),.05}$
All plugs	17		16.8637	356.0	71.4	1.8	3.1**				
(TC, LB; MB*, FH*)		2	1.9984	354.9	72.0	2.3	4.0	.02474	.01653	1.50	3.30

B = number of sites.
$N(P_1)$ ((P_0)) number of cores (plugs) ((polarity groups)).
R = resultant of direction cosines.
D (I) = declination (inclination) of mean remanence vector by intrusive (subscript i), polarity group (q), and all plugs (a) in degrees.
δ = angular standard error with subscripts as above in degrees.
α_{95} = radius of the cone of 95% confidence in degrees.
$\delta^2_b(\delta^2_w)$ = between-site (within-site) angular variance.
F = theoretical statistic for the .05 probability level.
*reverse polarities are normalized.
**α_{95} = 3.3° in Fisherian statistics.

the calculated value of δ^2_b/δ^2_w is greater than the corresponding statistics $F_{2(B-1),2(N-B),.05}$, so that the variance ratio test indicates that the site mean remanence directions within each plug are significantly distinct at the 95% confidence level. This result shows that even very small intrusive bodies such as these four plugs need to be adequately sampled as done here to derive a reliable intrusive mean remanence direction.

The variance ratio test was applied to the two plugs with normal remanence polarity giving each site unit weight. The calculated value of δ^2_b/δ^2_w (1.30) is considerably less than the statistic $F_{2,8,.05}$ (4.46). This result indicates that the intrusive mean remanence directions of Tin Cup Mountain and Lone Butte plugs are not significantly distinct, which suggests that they may have cooled simultaneously.

Applying the variance ratio test to the two plugs with reverse polarity gives a calculated value for δ^2_b/δ^2_w (19.87), which is considerably greater than the statistic $F_{2,18,.05}$ (3.55). This result suggests that the Mount Begbie and Forestry Hill plugs cooled at different times.

Finally, the calculated value of δ^2_b/δ^2_w (1.50) obtained by comparing the normally polarized sites to the reversely polarized sites is considerably smaller than the statistic $F_{2,30,.05}$ (3.30), which suggests that the normally polarized plugs as a group were intruded and cooled at essentially the same time as the reversely polarized plugs.

Discussion of Results

Translating the above statistical analysis into geological terms, several conclusions emerge. First, Tin Cup Mountain and Lone Butte plugs were probably intruded at the same time during one normal polarity interval of the Earth's magnetic field. However, as Lone Butte is represented by only two sites, the conclusion that the two plugs are comagmatic must be regarded cautiously. Second, Mount Begbie and Forestry Hill plugs were intruded either at different times during one reverse-polarity interval or during two different reverse-polarity intervals. The latter alternative seems more likely because these are intrusive bodies which probably cooled over a significantly long period of time (10^3) years to average out the effects of secular variation on remanence direction. Two facts support this contention: (a) the normalized site mean remanence directions are statistically significantly different within each plug, and (b) the angular dispersion of the normalized site mean remanence direction is 7.5°, which agrees closely with the angular dispersion values of 7° to 8° determined by Creer (1962) for the present Earth's geomagnetic field in the northern hemisphere. Therefore, three polarity intervals are probably represented, although two or four are possible. Third, the overall close grouping of the normalized site mean remanence directions (δ_a = 1.8°) and the fact that the normally polarized plugs are not statistically significantly different from the normalized reversely polarized plugs suggests that these plugs were intruded over a very short period of geologic time. Based on known and postulated rates of polarity reversal of the Earth's magnetic field during the past 10.6 m.y. from radiometrically dated lavas and sea-floor spreading rates, Cox (1968) has found the mean length of the polarity intervals to be 0.22 m.y. Therefore, it is possible that these plugs were intruded over a period of time in the order of 4×10^5 to 2×10^6 years, with 1×10^6 years being a reasonable compromise.

The geomagnetic pole position derived from the mean of the site mean remanence directions by assuming a magnetic geocentric dipole field during intrusion in late Tertiary times is 146.7° W, 84.9° N, with an ellipse of 95% confidence having semi-axes of 4.8° and 5.5° (Fig. 6).[1] This ellipse just includes the position of the Earth's present geographic pole, indicating that intrusion occurred during the late Tertiary or Holocene, and excludes the Earth's present geomagnetic pole (101.0° W, 75.4° N), which indicated the isolation of the stable primary thermoremanent magnetization (TRM). This pole position is in almost perfect coincidence with the pole determined for the surrounding late Miocene plateau basalts of 140.3° W, 84.0° N (δ_p = 4.8°, δ_m = 5.4°) (Fig. 6), based on 251 cores from 49 flows giving each flow unit weight (Symons, in press).

Statistically, the remanence directions of the plugs and flows form nearly identical statistical populations, which yields the geological impli-

[1]The semi-axes are 5.0° and 5.8° respectively in Fisherian statistics.

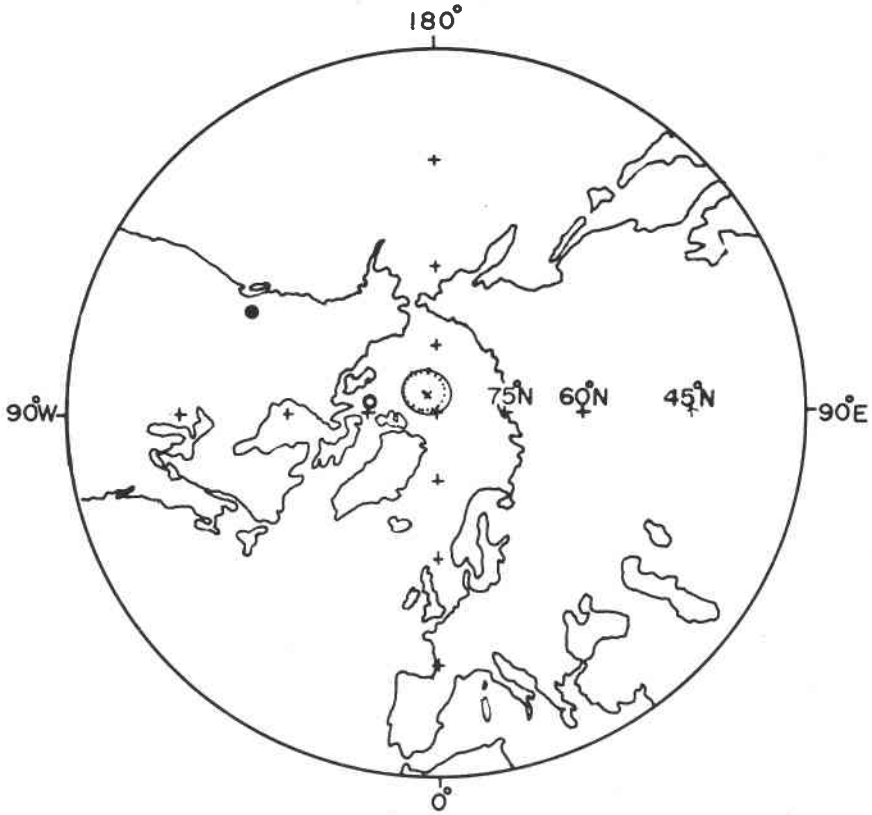


FIG. 6. Pole positions shown in the northern hemisphere north of latitude 30° N. The pole position for the gabbro plugs is shown by an × cross and its oval of 95% confidence by a solid line, and for the late Miocene plateau lavas by a dot and its oval of 95% confidence by a dotted line. The location of the sampling site is shown by a solid circle, of the present geomagnetic pole by an open circle, and of the geographic pole by the central cross.

cation that they are comagmatic. Thus, the hypothesis of Farquharson (1963) that the plugs represent the original vents or volcanic centers for the flows is entirely consistent with the paleomagnetic evidence. Considering the areal extent of the flows, it is likely that many more such volcanic centers are present.

Conclusions

It is concluded from this study that: (1) small intrusive bodies such as these plugs require thorough sampling at several sites as done in this study to derive a statistically significant result; (2) these four olivine gabbro plugs were possibly emplaced over a short span of geologic time ($<1 \times 10^6$ years); and (3) these plugs are comagmatic with the surrounding late Miocene plateau basalts and represent the original volcanic vents for extrusion of the flows.

Acknowledgments

The author thanks the following members of the Geological Survey of Canada: Dr. H. W. Tipper for pointing out the problem; Mr. K. W. Christie, Dr. A. Larochelle, Mr. G. W. Pearce, Dr. E. J. Schwarz, and Dr. H. W. Tipper for constructive criticism of the manuscript; and Mr. G. N. Freda and Mr. B. A. G. Smith for help in measuring and collecting the samples respectively. The paper is published with the permission of the Director of the Geological Survey of Canada.

CAMPBELL, R. B. and TIPPER, H. W. 1966. Bonaparte River (NTS 92P). Geol. Surv. Can., Map 3-1966.

——— 1969. Bonaparte River map-area, British Columbia (92P). Unpublished report, on open file, Geol. Surv. Can., 164 pp.

COX, A. 1968. Lengths of geomagnetic polarity intervals. J. Geophys. Res., **73**, pp. 3247–3260.

CREER, K. M. 1962. The dispersion of the geomagnetic field due to secular variation and its determination for remote times from paleomagnetic data. J. Geophys. Res., **67**, pp. 3461–3476.

FARQUHARSON, R. B. 1963. The petrology of several late Tertiary gabbroic plugs in the south Cariboo region, British Columbia. Unpublished M.Sc. thesis, Dept. Geology, Univ. British Columbia, Vancouver, 71 pp.

GEOLOGICAL SOCIETY. 1964. Geological Society Phanerozoic time scale. Quart. J. Geol. Soc. London, **1205**, pp. 206–262.

LAROCHELLE, A. 1968. L'application de la statistique au paléomagnétisme. Geol. Surv. Can., Paper 68-59, 19 pp.

LAROCHELLE, A. and BLACK, R. F. 1965. The design and testing of an alternating-field demagnetizing apparatus. Can. J. Earth Sci., **2**, pp. 684–696.

LAROCHELLE, A. and CHRISTIE, K. W. 1967. An automatic 3-magnet or biastatic magnetometer. Geol. Surv. Can., Paper 67-28, 28 pp.

MATHEWS, W. H. 1963. Thirteen potassium-argon dates of Cenozoic volcanic rocks from British Columbia. Rept. No. 2, Dept. Geology, Univ. British Columbia, Vancouver, 16 pp.

MATHEWS, W. H. and ROUSE, G. E. 1963. Late Tertiary volcanic rocks and plant-bearing deposits in British Columbia. Bull. Geol. Soc. Amer., **74**, pp. 55–60.

SYMONS, D. T. A. (in press). Paleomagnetism of the late Miocene plateau basalts in the Cariboo region of British Columbia. Geol. Surv. Can., Paper.

TARLING, D. H. and SYMONS, D. T. A. 1967. A stability index of remanence in paleomagnetism. Geophys. J. Roy. Astron. Soc., **12**, pp. 443–448.